	SAMRAT ASHOK TECHNOLOGICAL INSTITUTE (DEGREE) VIDISHA (M.P)
Department	Mechanical EngineeringDepartment
Branch	Common to all 1st Year Branches
Year & Sem	B.Tech. First Year (I /IISemester)
Session	Batch 2026-27 onwards

Category of Course	Course Title	Course Code	L	T	P	Credits - 2
ESC	Workshop Practice	ME26102P	0	0	4	

Course Code	Course Title	Maximum Marks Allotted					Total Marks
		Theory Paper			Practical		
		End Sem.	Mid Sem. MST	Quiz/ Assignment	End Sem.	Lab Work	
ME26102P	Workshop Practice	--	--	--	50	50	100

Course Objectives	This Subjects deals with the basic knowledge related to production such as casting, welding, joining etc. After completing this subjects students are able to analyze the difference between various manufacturing techniques and their specific applications.
Prerequisite Knowledge	• Familiarity with workshop safety rules in different shops. • Knowledge of different systems of units and their inter-conversions. • Basic knowledge of different standards (ISO, BIS, DIN, OHSAS etc.) • Awareness of housekeeping and maintenance practices in workshops.
Course Description	Engineering Workshop Practice is a fundamental laboratory-based course designed to provide students with practical exposure to basic manufacturing and fabrication processes used in engineering industries. The course develops hands-on skills, technical knowledge, safety awareness, and familiarity with commonly used workshop tools, machines, and equipment. Students learn essential workshop operations through practical training in carpentry, foundry, blacksmithy, welding, and fitting shops. The course emphasizes the importance of workmanship, precision, material handling, and industrial safety while performing various manufacturing operations. It bridges the gap between theoretical engineering concepts and real industrial practices by enabling students to fabricate simple jobs and components using different workshop techniques.
Course Outcomes	**Upon completion of the course, student will be able to:** 1. Identify and understand the functions, construction, and safe handling procedures of various tools, machines, and equipment used in carpentry, foundry, blacksmithy, welding, and fitting shops. 2. Perform basic carpentry operations such as measuring, marking, cutting, planing, chiseling, drilling, and preparation of common wooden joints with proper workmanship. 3. Prepare moulds and carry out basic foundry operations including pattern making, mould preparation, core making, and simple casting processes while understanding casting defects and safety practices. 4. Demonstrate blacksmithy and forging operations such as upsetting, bending, tapering, and punching using forging tools and forge furnaces safely and efficiently. 5. Perform basic welding operations using arc and gas welding processes, prepare different welded joints, and apply proper welding safety measures and inspection techniques. 6. Carry out fitting shop operations such as filing, sawing, drilling, tapping, and assembly work using precision measuring instruments while maintaining dimensional accuracy and tolerances. 7. Interpret simple engineering drawings and apply workshop practices to fabricate basic engineering components and models. 8. Develop practical skills, teamwork, safety awareness, problem-solving ability, and understanding of manufacturing processes commonly used in engineering industries.

List of Experiments

1.	To prepare a T- Half lap joint in carpentry shop.	CO1, CO2
2.	To prepare a Dovetail joint in carpentry shop.	CO1,CO2
3.	To prepare a model of single door window frame in carpentry shop.	CO1, CO2
4.	To prepare chisel from given mild steel rod in black smithy shop.	CO1, CO4
5.	To prepare butt joint by electric arc welding process.	CO1, CO5
6.	To prepare lap joint by electric arc welding process.	CO1, CO5
7.	Demonstration of Gas Welding.	CO1, CO5
8.	Demonstration of Tungsten Inert Gas Welding (TIG).	CO1, CO5
9.	To fabrication of table frame in welding shop.	CO1, CO5
10.	To prepare sand mould single piece pattern in Foundry Shop.	CO1, CO3
11.	To prepare sand mould two-piece pattern in Foundry Shop.	CO1,CO3
12.	To prepare V Joint in fitting shop.	CO1,CO6
13.	To remove material small square section with dimensions from workpiece in fiiting shop.	CO1, CO6
14.	Assembly of simple engine/machine in fitting Shop.	CO1, CO6